Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First -Semester 2025-2026

Mid-Semester Test
(EC-2)

Course No.	: AIMLZG557
Course Title	: Artificial and Computational Intelligence
Nature of Exam	: Closed Book
Weightage	: 30%
Duration	: 2 Hours
Date of Exam	: XX/XX/XXXX (AN)

No. of Pages = 2
No. of Questions = 4

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q.1 a) Identify and explain two application domains of AI and discuss the challenges AI faces in these domains.

b) Differentiate between tree search and graph search. Provide an example scenario where graph search is more suitable.

c) Compare and contrast PSO and ACO as population-based optimization algorithms.

d) Define the transition probability in Ant Colony Optimization (ACO) and explain the role of parameters α (pheromone importance) and β (heuristic importance).

[2+2+2+2=8 Marks]

Q.2 A **warehouse robot** begins at C1, tasked with collecting Items (It) and reaching the Dock (DV) at C4 on a 4x4 grid. The robot must navigate hazards like blocked cells (#), Spills (SP) with high movement costs, while leveraging Energy packs (E) to reduce costs. Using the A* algorithm, the robot calculates the optimal path by considering both actual path cost and a heuristic. The agent is equipped with four distinct actions: Move Up, Move Down, Move Left, and Move Right. However, it cannot move into blocked cells.
The warehouse is represented as a **4×4 grid** below:

	1	2	3	4
A	It	E	#	SP
B	SP	It	E	SP
C	S	E	It	DV
D	It	SP	SP	E

Entering **SP**: +12; collecting **It**: −8 (reward); energy pack E: +1.
Heuristic. h(n) = Manhattan(robot, DV) + (# remaining Items) + (# adjacent SP to robot in resultant state).

a) Expand and depict the search tree for exactly 2 levels. (Level 0 = Start).
b) Calculate Path costs and heuristics (g/h/f) for all generated nodes.
c) Apply IDA* search and show the first 3 CLOSED list updates with OPEN/CLOSED status.

[3+3+3 = 9 Marks]

Q.3 A large electronics manufacturing facility wants to ensure that only defect-free components are shipped to customers. An **Automated Quality Control System** is being designed with AI-enabled sensors and actuators. The system inspects each component on the conveyor belt, identifies defects by comparing them with a database of standards, and classifies them into either **Accept (Pass)** or **Reject (Fail)** categories.

System Details:

- Sensors are installed at critical checkpoints along the production line to detect cracks, misalignments, missing solder points, and irregular voltages.
- Defective components are automatically diverted to the **Reject bin** using conveyor diverters, while non-defective ones proceed to **Packaging**.
- The system must also maintain logs and raise alerts if defect rates exceed a threshold.

a. Provide the complete problem formulation.
b. Provide the PEAS description.
c. Identify the various dimensions of the task environment with appropriate justification foreach in no more than 30 words. **[2+2+2=6 Marks]**

Q.4 A hospital must create a **weekly schedule** for 5 nurses (N1–N5) across **three shifts per day** (Morning, Evening, Night). The schedule must ensure that all shifts are covered while preventing overwork. A nurse cannot be assigned **more than 5 shifts per week (Mon-Fri)**.

You are tasked with designing a **Genetic Algorithm** to generate feasible schedules. Each chromosome encodes the assignment of shifts to nurses as a binary string of length 15 (5 nurses × 3 shifts). The fitness function rewards full coverage and penalizes uncovered shifts and overworked nurses.

a) Represent 4 random chromosome states with their schedules and calculate the **fitness score** for each.
b) Describe the **Selection, Crossover, and Mutation** process for this problem. Demonstrate one full iteration using a tournament selection of 2, a 1-point crossover, and a single bit-flip mutation. Clearly show the generated offspring and their fitness.
[3+4=7 Marks]

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